

A Fictional Lens for Imaginary Telescopes

We describe a fictional lens that exists only in this sample file. It is used to test how the reader splits a paper into sections and sentences. Nothing here refers to a real instrument.

1 INTRODUCTION

Imaginary telescopes need imaginary lenses (Smith et al., 2020). This paper walks through the design of one such lens, e.g. the choice of glass and the size of the pupil. The lens is 3.5 mm thick and weighs 12 g.

Section 2 lists the requirements. Section 3 describes the design, and Section 4 closes the paper.

2 REQUIREMENTS

The lens must survive a launch that never happens [3]. It must also focus light onto a detector that was never built.

2.1 Optical Quality

The wavefront error must stay below 20 nm RMS across the field (Lee and Park, 2019a; Kim, 2021). We verify this with a made-up test bench.

2.2 Mass and Volume Limits for a Very Small Imaginary Spacecraft

The whole assembly must fit in 1U. Fig. 1 shows the layout that satisfies this limit.

FIGURE 1 | The layout of the fictional lens. This caption should not be read aloud.

3 DESIGN

We chose a single aspheric surface. Dr. Jones suggested this approach in a conversation that did not take place. The resulting design meets every requirement listed above.

4 CONCLUSION

A fictional lens can be designed in a few paragraphs. Real lenses take longer.

ACKNOWLEDGMENTS

We thank nobody in particular.

REFERENCES

Kim, A. (2021). An imaginary paper. J. Fict. Opt. 1, 1-2.

